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Predicting the value of football players: machine learning techniques and sensitivity analysis based on FIFA and real-world statistical datasets

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Abstract
The study focuses on applying machine learning methodologies to football player data for predicting player market values in the dynamic football market. Player datasets are rich, encompassing performance metrics, physiological attributes, and contextual variables. Machine learning models, including both traditional and advanced methods, effectively extract insights from complex data to estimate player market values. Addressing challenges like overfitting and computational complexity involves applying regularization, feature engineering, and interpretability tools to manage high-dimensional data and improve predictive accuracy. In this study sensitivity of selected models (Support Vector Regression (SVR), Random Forest Regression (RFR), Extreme Gradient Boosting (XGB), and Categorical Boosting (CAT)) models to extracted data from FIFA 19 and Real-world Statistical Datasets evaluated by Shapley Additive Explanations (SHAP) and the 20 most relevant features selected in the ranking of SHAP for each regression model. Then, models optimized with two meta-heuristic algorithms demonstrated their performance in predicting the market values of players. Dempster-Shafer Theory (DST) was utilized to develop an ensemble of models to overcome overfitting problems, and Fourier amplitude sensitivity testing (FAST) gave insight for future data extractions. The analysis of market values for players revealed significant model performance variations. XGSC hybrid model demonstrated exceptional precision with a minimal error of 1.7 million dollars (10% of average measured value), followed by RSCX_SC with misestimations of 2 million dollars (13.3% of average measured value). Extracted results suggested that models, especially ensemble form, offer reliable accuracy for club managers and stakeholders, aiding in strategic player selection based on previous performance. This approach proves particularly beneficial for optimizing player salaries, especially when considering a prominent team with market values above average.

Keywords Football transfer market · Hybrid machine learning model · Ensemble modelling · Dempster-Shafer theory · Fourier amplitude sensitivity testing

Abbreviations

RFR	Random Forest Regression
RF	Random Forest
R_n	Training dataset with n data points
X	Input vector
m	Number of features
Y	Single-valued output
$\widehat{H}$	Predictive function
Θ_K	Randomly generated vectors
$\widehat{Y}$	Final predicted value
n_{tree}	Number of trees
m_{try}	The number of randomly selected attributes for each tree division
SVR	Support Vector Regression
ϵ	Noise parameter
y_i	Dependent variable
$\langle w, x_i \rangle$	The dot product of the weight Vector and feature vector
b	Error term added in
CAT	Categorical Boosting
XGB	Extreme Gradient Boosting
DS	Dataset
k	Variable that specifies the number of trees
f_k	$k-th$ tree

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Ω	The parameter presenting the complexity of the model
w	The weight of each leaf
T	The total number of leaves
t	Iteration number
HGSO	Hunger Games Search Optimization
$\overrightarrow{X(t)}$	Individual positions
$\vec{X}_b$	The best individual's position
$\overrightarrow{W}_1$	Hunger weight
$\overrightarrow{W}_2$	Hunger weight
r_1	Random number
r_2	Random number
l	Sensitivity control variable
E	Variation control parameter
$F(i)$	Cost function value for each population
BF	Best cost function value achieved in the current iteration
$Sech$	Hyperbolic function
Max_{iter}	Upper limit of iterations
N	Population size
r_3	Random number
r_4	Random number
r_5	Random number
$AllFitness(i)$.	Cost function value of each population in the current iteration
WF	Worst fitness
BF	Best fitness
r_6	Random number
SCSO	Sand Cat Swarm Optimization
$\vec{X}$	Positional vector of the search agent
$\vec{X}_b(t)$	The location of the leading contender during iteration t
$\vec{X}_c(t)$	The most recent location of the hunting agent at repetition t
$\vec{r}_G$	Overall sensitivity span
$iter_c$	The current iteration
θ	Arbitrary angle between 0 and 360 degrees
$\vec{X}_{md}$	A position formed by combining the optimal and current positions
SHAP	Shapley Additive Explanations
z	Binary vector
φ_i	Feature attribution value
F	The non-zero set of inputs in z
S	The subset of F with the ith feature excluded
R^2	Coefficient of determination
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
NMSE	Normalized Mean Square Error
PI	Prediction Interval
$\underline{P}_i$	Predicted market values
P	Average of all predicted values
M_i	Real market values
$\bar{M}$	An average of all real values
k^2	Standardized error value
$t_{(\alpha/2,\ N-2)}$	t-value for the desired level of confidence (α) and the degrees of freedom ($N-2$)

1 Introduction

Soccer is an adored team sport played globally with two teams of eleven players each. Its popularity is evident through major competitions such as the FIFA World Cup and various domestic leagues, contributing to its widespread appeal [1]. Football has evolved from a simple sport into a social phenomenon, as seen in the considerable media coverage and attention it receives [2]. The market value of football players is a dynamic and multifaceted concept determined by factors such as skill level, age, performance, contractual situation, and demand from potential buyers [3]. Transfer fees are negotiated based on these considerations, with top-tier players often commanding significant sums [4]. External factors, including industry trends, economic conditions, and the financial capabilities of clubs, also influence player values [5]. Additionally, commercial appeal, endorsements, sponsorships, and image rights contribute to players' overall value as both athletes and brands. In essence, the market value of football players is a complex and ever-changing aspect of the sport's economic context [6].

Player value evaluation in football has multiple purposes, including evaluating club assets for financial statements. This process, involving various stakeholders, becomes vital when clubs consider recruiting a player under contract with another team, leading to the determination of an asking price. In these states where a player's contract is expiring, clubs must decide whether to renew or allocate the freed-up capital elsewhere. In essence, player valuation acts as a decision-making tool for clubs, influencing choices in recruitment and investment [7–9]. Furthermore, in a highly interconnected market, it is essential for representatives of players to evaluate the worth that a club assigns to the athlete when engaging in salary negotiations. The greater the perceived value of the player, the increased ability of representatives to negotiate a higher salary [10]. Researchers can assess players' value through either actual transfer fees, representing market prices rarely disclosed publicly, or by utilizing media companies like Transfermarkt.de, Marca, or Kicker.de, offering market values based on expert judgment or community input [11, 12].

2 Literature review

Traditionally, player value estimation has been reliant on subjective assessments, expert opinions, and statistical analysis [13]. While these methods offer valuable insights, they may not possess the same depth and predictive power

as Machine Learning (ML) algorithms [14]. Exploring the connection between ML and the market value of football players is a fascinating area within the football industry. ML models, by uncovering intricate patterns and relationships in the data, have the potential to offer more precise and unbiased player valuations, surpassing conventional subjective assessments and statistical analyses [15]. ML can simultaneously take into account a diverse array of factors, encompassing player performance metrics, age, experience, contractual standing, market trends, and contextual variables, for the purpose of predicting a player's market value. This all-encompassing methodology facilitates a more thorough and discriminating comprehension of a player's value in the transfer market [16]. Besides, ML models excel in recognizing undervalued players, forecasting potential performance trends, and discerning the influence of different aspects on a player's value in the market, thereby offering valuable insights for football clubs and stakeholders [17]. The adoption of ML in predicting the market value of football players brings forth numerous advantages, including heightened predictive accuracy, the capability to optimize transfer strategies and financial investments, and the potential to uncover market inefficiencies and prospects. By applying insights driven by data, clubs can elevate their decision-making processes, fine-tune squad composition, and navigate the complexities of the transfer market with increased precision and strategic foresight [18–20].

The proficiency of ML in evaluating players may be based on a diverse range of comprehensive datasets (covering player statistics, match performance, injury history, and contextual factors) related to a special league or football season. For instance, McHale and Holmes [21] suggested the use of the Extreme Gradient Boosting (XGB) model for assessing the transfer fees of renowned professional footballers such as Lionel Messi, Cristiano Ronaldo, Neymar, Jan Oblak, and others. Wang et al. [22] examined EPL transfer fees using a Nash equilibrium model and various dynamic systems, suggesting neural networks for optimal value determination. Lee et al. [23] employed ensemble models, considering 124 features, to estimate players' market values, evaluating them based on various statistical metrics. Meanwhile, Al Asadi and Taşdemir [24] aimed to estimate players' market values using four regression models, Multiple Linear Regression (MLR), Linear Regression (LR), Decision Trees (DT), and Random Forest Regression (RFR). Considering another set of models, Behravan and Razavi [14] introduced a hybrid form of Support Vector Regression (SVR) optimized with particle swarm optimization (PSO) and opened a new insight into the application of novel ML models in market value prediction.

3 Research gaps and study overview

Although a vast variety of datasets from video games like FIFA and Football Manager (FM) and methods have been utilized for predictions in other fields such as football results prediction and football players' Injury recovery time prediction, for example, predicting Champions' League group stage matches using Naive Bayes, LogitBoost, Bayesian networks, Artificial Neural Networks, K-Nearest Neighbors, and Random Forest [25] or for specific teams, such as predicting Tottenham Hotspur's results in the 95/96 and 96/97 Premier League seasons [26] and utilizing the PubMed database, employing tree-based ensemble methods, Artificial Neural Networks, and Support Vector Machines to predict injuries in elite youth soccer players [27]. But there is a gap in the literature related to (1) utilizing the reach dataset extracted from reliable sources such as sofifa.com, which integrates real-world statistics, and the FIFA game dataset, which contains a broad range of features from players characteristics and performance to social factors such as international reputation (described in Section 5) (2) utilizing novel and robust feature selection methods such as shapley additive explanations (SHAP) which prepares visual representation of features importance (described in Section 5) (3) utilizing state of art hybrid and ensemble methods to develop prediction frameworks based on more precise and recently developed base models and optimizers which became an inspiration for development of this research. The significance of predicting the market value of football players through a comprehensive analysis of players' characteristics, performance, and international reputation, utilizing both FIFA19 data and real-world statistics, is vast and diverse. This predictive approach not only combines virtual player representations with actual performance metrics but also is a tool for various stakeholders in the football industry. It facilitates informed decision-making in player valuations and transfers, allowing clubs to understand a player's potential and mitigate risks associated with high-value transfers. Additionally, it aids in talent identification and strategically hiring players based on specific requirements. Moreover, it opens up marketing and commercial opportunities by enhancing a club's brand value and engaging fans globally.

Besides four base regression models, which are introduced in subsections 4.1 to 4.4, in this study, two more recent optimization algorithms (Sand Cat Swarm Optimization and Hunger Game Search Optimization) are selected for fine-tuning parameters of SVR, RFR, XGB, and CAT models, to estimate the market value of players from top-tier European leagues based on vast range influential factors. These algorithms (as introduced in

Section 4.5 and 4.6) address drawbacks of traditional optimization algorithms, such as limited scope, overfitting, and computational inefficiency, by offering enhanced exploration and exploitation of the solution space, improving prediction accuracy, and increasing computational efficiency. Of course, this article has gone further by introducing an ensemble form of four models utilizing Dempster-Shafer theory (DST) and optimizing that with Sand Cat Swarm Optimization for more reliable and accurate predictions, as described in Section 6. The reliability and compatibility of the developed ensemble prediction framework assume paramount importance in the evaluation of football players, as club managers increasingly rely on contemporary datasets to inform decisions regarding contract negotiations, player recruitment strategies, and financial planning for forthcoming seasons. This framework, deeply rooted in sports analytics, represents an innovative application of statistical methodologies and data science in the context of football management as it involves the systematic analysis of extensive datasets to extract actionable insights and optimize decision-making processes.

4 Regression and optimization algorithms

Among others, for this prediction task, the base models are: Random Forest Regression (RFR), Decision Tree Regression (DTR), Support Vector Regression (SVR), and Extreme Gradient Boosting (XGB), since they have proved to be effective in handling high-dimensional data in previous tasks and due to their excellent performance for different regression tasks. The reasons for valuing RFR and XGB are their high accuracy and robustness by ensemble learning and boosting techniques, and kernel methods of SVR are effective in modeling nonlinear relationships. DTR offers simplicity and interpretability; hence, it is useful to understand factor contributions despite susceptibility to overfitting. These models and two new optimization algorithms will be described in detail in the following subsections:

4.1 Random Forest Regression (RFR)

Random Forest (RF) is a potent regression algorithm utilizing multiple independent decision trees. Decision Trees, such as Classification and Regression Trees, introduced by Breiman et al. [28], are influenced by the learning process and input data complexity. Trees consist of decision and leaf nodes, where the input vector X with m features yields a single-valued output Y [29, 30]. The training dataset with n data points is denoted as R_n in Eq. (1):

$$R_n = \left\{\left(X_1, Y_1\right),\ \left(X_2, Y_2\right),\ \ldots\ ,\ \left(X_n, Y_n\right)\right\},\ X \in R^m,\ Y \in R. \tag{1}$$

In the training phase, the algorithm strategically divides input data to create decision trees until reaching terminal nodes. A predictive function $\widehat{H} = (X, R_n)$ is then used for predictions on the training dataset R_n. In Random Forest Regression (RFR), L tree-structured base classifiers denoted as $H = (X, \Theta_K)$ are employed, using randomly generated vectors Θ_K [31] for each decision tree. The creation of a RF involves randomly selecting feature or training dataset subsets for each tree.

The Bootstrap technique creates a sample from the training dataset by randomly selecting n observations with replacement. The bagging algorithm uses multiple bootstrap samples, applies $(S_n^{\Theta_1}, \ldots, S_n^{\Theta_q})$ a tree decision algorithm to generate q prediction trees and organize $\widehat{h}\left(X, S_n^{\Theta_1}\right)$, ., $\widehat{h}\left(X, S_n^{\Theta_q}\right)$ into an ensemble. Each tree produces an output value, resulting in a sequence of predictions $\widehat{Y}_1 = \widehat{h}\left(X, S_n^{\Theta_1}\right),\ \widehat{Y}_2 = \widehat{h}\left(X, S_n^{\Theta_2}\right), \ldots, \widehat{Y}_q = \widehat{h}\left(X, S_n^{\Theta_q}\right)$. The aggregation process calculates the mean value of these predictions, yielding the final predicted value $\widehat{Y}$ [32].

$$\widehat{Y} = \frac{1}{q}\sum_{l=1}^{q}\widehat{Y}_l = \frac{1}{q}\sum_{l=1}^{q}\widehat{H}\left(X, R_n^{\Theta_l}\right) \tag{2}$$

$\widehat{Y}_l$ signifies the output of the l_{th} tree, with l reaching from 1 to q.

The bagging technique in RF training improves model stability by allowing some data points to be used multiple times. RF intentionally avoids tree pruning during construction, ensuring computational efficiency. The RF regression method is known for its simplicity, requiring fine-tuning of only two parameters: the number of trees n_{tree} and the number of randomly selected attributes for each tree division m_{try} [33]. More trees in a forest typically improve model accuracy and robustness but can increase computational complexity. The benefit diminishes after a certain point, often requiring $n_{tree} = 500$ for predictions. This highlights each decision tree's unique strength and their interdependence. Conversely, increasing m_{try} enhances individual tree strength but raises inter-tree correlation [34]. The fundamental steps of the RF are as follows:

> **Bootstrap sampling** creates n_{tree} subsets from the original dataset. **Random feature** selection in unpruned regression trees involves choosing a subset of m_{try} predictors at each node split. **Ensemble prediction** aggregates predictions from the n_{tree} sub-models, using majority voting in classification and averaging in regression for improved overall accuracy and reliability.

The RF methodology is improved in predictive modeling through steps involving the creation of regression trees using bootstrap samples and out-of-bag (OOB)

samples. Two-thirds of the data are used for training, and one-third is reserved for OOB evaluation. Figure 1 presents a schematic representation of RFR [35]. This figure illustrates the structure and functionality of the RFR algorithm. It shows how multiple decision trees are built using subsets of the training data and features. The figure demonstrates how these trees aggregate their predictions to provide a final output. The logical relationship here is that the combined output of multiple trees improves prediction accuracy and robustness compared to individual decision trees. The run time of a single RFR model in this study was 1.516 s.

4.2 Support Vector Regression (SVR)

While traditional regression models focus on minimizing squared error to estimate coefficients, Support Vector Regression (SVR) is designed for predicting continuous values based on the principle of structural risk minimization [36]. SVR aims to find a function that approximates the data within a specified error margin (ϵ) while maintaining a flat model complexity to prevent overfitting [37].

$$minimize \frac{1}{2} \| w \|^2 \tag{3}$$

$$Subject\ to \begin{cases} y_i - \langle w, x_i \rangle - b \leq \epsilon \\ \langle w, x_i \rangle + b - y_i \leq \epsilon \end{cases} \tag{4}$$

In SVR, the objective is to minimize$\| w \|^2$, which ensures that the decision boundary should be as flat as possible. In the expression, y_i is the dependent variable, $\langle w, x_i \rangle$ is the dot product of the weight vector and feature vector, b is the error term added in, and ϵ is the noise parameter. ϵ is the tolerated error, the plausible error [38].

Figure 2 presents a schematic representation of SVR. Figure 2 details the SVR model, highlighting how SVR constructs a regression function that fits within a specified margin while minimizing the error. It visually represents the concept of maximizing the margin between support vectors and the regression function, thus ensuring better generalization. The run time of a single SVR model in this study was 1.158 s.

4.3 Categorical boosting (CAT)

Categorical Boosting (CAT) is an advanced gradient boosting algorithm built upon decision trees, offering notable improvements in categorical classification while being computationally more efficient than other similar algorithms [39]. Besides, it has other main strengths: effective vector representation of categorical data due to the idea of ordered boosting [40]. Hence, it is used for response coding, where for a categorical feature, the mean target values of the corresponding data points are used to encode it, making the model better in its prediction. The run time of a single CAT model in this study was 2.326 s.

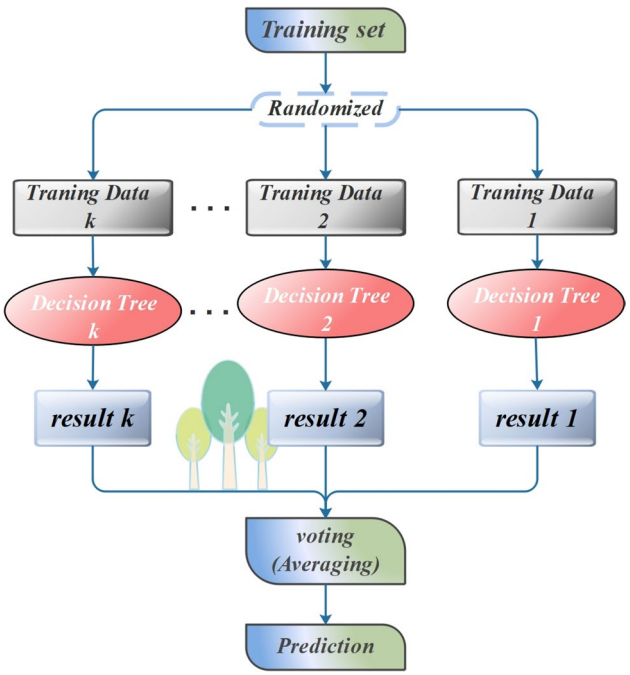


Fig. 1 The flowchart of Random Forest Regression (RFR)

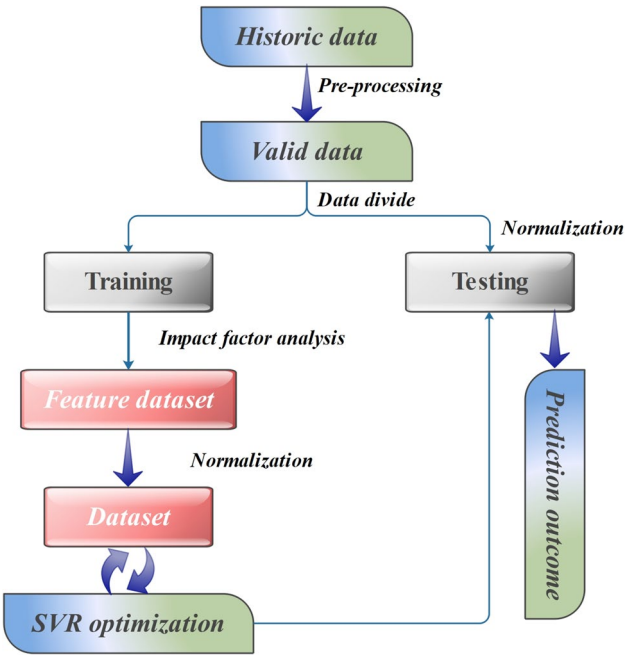


Fig. 2 The flowchart of Support Vector Regression (SVR)

4.4 Extreme Gradient Boosting (XGB)

Extreme Gradient Boosting (XGB) is a popular choice due to high execution speed, attributed to its computation capabilities [41]. Operation of the XGB is in the following manner:

Suppose we have a dataset DS with m features and n examples, where $DS = \{(X_i, y_i) : i = 1 \ldots n,\ x_i \in R^m,\ y_i \in R\}$. Then, the predicted value, $\hat{y}_i$of an ensemble tree model will be generated as follows:

$$A_i = \varnothing\left(X_i\right) = \sum_{k=1}^{K} f_k\left(X_i\right),\ f_k \in F \tag{5}$$

where k specifies the number of trees and f_k indicates the $(k - th$ tree).

To address the aforementioned equation, the essential step involves minimizing both the loss and regularization objectives by identifying the optimal set of functions.

$$\varphi(\varnothing) = \sum_{i} 1\left(y_i, A_{\cdot i}\right) + \sum_{k} \Omega\left(f_k\right) \tag{6}$$

The loss function is defined as the difference between the predicted output, $\hat{y}_i$, and the actual output, y_i. Ω gauges the complexity of the model, which helps prevent overfitting:

$$\Omega\left(f_k\right) = \gamma\ T + \frac{1}{2}\lambda \parallel w \parallel^2 \tag{7}$$

where w denotes the weight related to each leaf, and T is the count of all leaves.

During model training, boosting is utilized with decision trees to decrease the objective function reaching the minimum value by introducing a new function f as the model iteratively learns. Consequently, in each iteration (t), a new function (tree) is incorporated into the model.

$$\varphi^{(\mathrm{t})} = \sum_{i=1}^{n} 1(y_i, A_{\cdot i}^{(t-1)} + f_t\left(X_i\right)) + \Omega\left(f_t\right) \tag{8}$$

$$\Omega_{\text{split}} = \frac{1}{2}\left[\frac{\left(\sum_{i\in I_L} g_i\right)^2}{\sum_{i\in I_L} h_i + \lambda} + \frac{\left(\sum_{i\in I_R} g_i\right)^2}{\sum_{i\in I_R} h_i + \lambda} - \frac{\left(\sum_{i\in I} g_i\right)^2}{\sum_{i\in I} h_i + \lambda}\right] \tag{9}$$

$$\mathrm{g_i} = \partial_{A^{t-1}} 1(y_i,\ A.^{(t-1)}) \tag{10}$$

$$h_{\mathrm{i}} = \partial^{\,2}_{A^{t-1}} 1(y_i,\ A.^{(t-1)}) \tag{11}$$

The run time of single XGB model in this study was 1.230 s.

4.5 Hunger games search optimization (HGSO)

Yang et al. [42] introduced the HGSO algorithm in 2021, specifically designed for addressing continuous optimization problems. It is inspired by animal behaviors related to fear and hunger.

4.5.1 Approach food

This subsection introduces a mathematical model for hunger-driven behavior, with Eq. (12) describing the following hunger and individual supportive communication actions, simulating the contraction mode.

$$\vec{X}(t+1) = \begin{cases} \overrightarrow{X(t)}.(1 + randn(1)), & r_1 < 1 \\ \overrightarrow{W}_1.\vec{X}_b + \vec{R}.\overrightarrow{W}_2.\left|\vec{X}_b - \overrightarrow{X(t)}\right|, & r_1 > 1,\ r_2 > E \\ \overrightarrow{W}_1.\vec{X}_b - \vec{R}.\overrightarrow{W}_2.\left|\vec{X}_b - \overrightarrow{X(t)}\right|, & r_1 > 1,\ r_2 < E \end{cases} \tag{12}$$

$\overrightarrow{X(t)}$ represents individual positions, $\vec{X}_b$ is the best individual's position, $\overrightarrow{W}_1$ and $\overrightarrow{W}_2$ are hunger weights, $\vec{R}$ ranges from $[-a,\ a]$,r_1 and r_2 are randomly selected numbers between [0, 1], $randn(1)$ is a normally distributed random number, t is the current iteration, l is the sensitivity control variable in the HGSO algorithm, and E signifies variation control in all positions Eq. (13):

$$|E = \mathrm{sech}(|F(i) - BF|)| \tag{13}$$

The value of cost function for each population is represented by $F(i)$, i ranges from 1 to n. Sech represents the hyperbolic function, equivalent to $(\mathrm{sech(x)} = \frac{2}{e^x + e^{-x}})$. Equation (14) provides the expression for $\vec{R}.BF$ stands for the best cost function value achieved in the current iteration.

$$\vec{R} = 2 \times\ a \times\ rand - a \tag{14}$$

$$a = 2 \times \left(1 - \frac{t}{Max_{iter}}\right) \tag{15}$$

Max_{iter} represents the upper limit of iterations, and rand represents a randomly selected number within the range [0, 1].

4.5.2 Hunger role

This subsection mathematically formulates the all individuals' hunger behavior throughout the search and the equation for $\overrightarrow{W}_1$ is provided in Eq. (16):

$$\overrightarrow{W_1(i)} = \begin{cases} hungry(i).\frac{N}{SHungry} \times r_4, & r_3 < | \\ 1 & r_3 > | \end{cases} \quad (16)$$

The equation for $\overrightarrow{W_2}$ is detailed in Eq. (17):

$$\overrightarrow{W_2(i)} = (1 - \exp(-|hungry(i) - SHungry|)) \times r_5 \times 2 \quad (17)$$

N is the population size, *hungry* indicates the starvation level of each population, *SHungry* is the cumulative starvation feeling across all populations, and r_3, r_4, and r_5 are randomly selected numbers between [0, 1]. The mathematical representation of each population' starvation is expressed in Eq. (18):

$$hungry\,(i) = \begin{cases} 0, & AllFitness(i) == BF \\ hungry(i) + H, & AllFitness(i)! = BF \end{cases} \quad (18)$$

The cost function value of each population in the current iteration is represented by $AllFitness(i)$. A supplementary starvation value, denoted as H, is introduced based on the standing starvation. H is specified in Eq. (19):

$$H = \begin{cases} LH \times (1 + r), & TH < LH \\ TH, & TH \geq LH \end{cases} \quad (19)$$

$$TH = \frac{F(i) - BF}{WF - BF} \times r_6 \times 2 \times (UB - LB) \quad (20)$$

H is limited by a lower bound, LH, with r as a randomly selected number [0, 1]. WF and BF represent the worst and the best fitness in the current iteration, respectively. $F(i)$is the fitness of each population, and r_6 is a random number [0, 1]. LB and UB are the lower and upper bounds of dimensions. Algorithm 1 provides the pseudocode related to the HGSO algorithm:

Algorithm 1 Pseudocode of HGSO Algorithm

Set up the variables, including $N, Max - iter, l, D$ *and SHungry.*
Set the initial positions of the individuals $X_i(i = 1,2, \ldots, N)$
While $(t \leq Max - iter)$
Calculate the cost function value for each population.
Inform BF, WF, X_b, *and BI*
Discovery of the Hungry, W_1, *and* W_2 *using Eqs.* (14), (15) *and* (16), *respectively.*
For *each population*
Find E by Eq. (12):
Update $\boldsymbol{R}$ *and positions by Eqs.* (13) *and* (11)
End For
$t = t + 1$
End While
Return BF *and* X_b

The run time of the SVHG, RFHG, CAHG, and XGHG hybrid models developed by optimizing four models of SVR, RFR, CAT, and XGB by HGSO were 4.600, 6.040, 9.280, and 4.920 s respectively.

4.6 Sand cat swarm optimization (SCSO)

The inspiration of this optimization algorithm is drown from the foraging behavior of sand cats in desert environments

[43], emphasizing cats' ability to locate prey through low-frequency sounds. It centers on determining the optimal point within an exploration area, similar to the cat's natural hunting instincts. The algorithm employs a search agent that systematically explores the area, updating its location to converge toward the estimated optimal value [44]. Comprising a prey-locating along with prey-capturing mechanism, the SCSO algorithm mimics the hunting patterns of sand cats using a mathematical equation to guide population search patterns for optimization and solution identification within the exploration area.

$$\vec{X}(t+1) = \vec{r}.\,\vec{X}_b(t) - rand(0,1).\,\vec{X}_c(t) \tag{21}$$

$\vec{X}$ represents the positional vector of the search agent, while t represents the iteration number for the current cycle. $\vec{X}_b(t)$ is the location of the leading contender during iteration t, and $\vec{X}_c(t)$ represents the most recent location of the hunting agent at repetition t. The variable r represents the sensitivity range of sand cats to low-frequency sounds. This sensitivity is characterized as follows:

$$\vec{r} = \vec{r}_G \times rand(0,1) \tag{22}$$

$\vec{r}_G$ shows the overall sensitivity span, which reduces linearly from 2 *to* 0. Equation (23) may be explained upon as follows:

$$\vec{r}_G = s_M - \left(\frac{s_M \times iter_c}{iter_{max}}\right) \tag{23}$$

$iter_c$ represents the current iteration, while $iter_{max}$ represents the total number of iterations. Additionally, given that sand cats identify low frequencies at $2kHz$, the value of S_M is adjusted to 2. The procedure of this algorithm initiates the predator assault stage after completing the prey exploration, and the predator attack mechanism of the sand cats' population can be explained as follows:

$$\vec{X}_{md} = \left|rand(0,1).\vec{X}_b(t) - \vec{X}_c(t)\right| \tag{24}$$

$$\vec{X}(t+1) = \vec{X}_b(t) - \vec{r}.\vec{X}_b(t).cos(\theta) \tag{25}$$

θ is an arbitrary angle between 0 and 360 degrees, where $cos(\theta)$ yields values from -1 to 1. $\vec{X}_{md}$ represents a position formed by combining the optimal and current positions.

Sand cats navigate unique circular paths by choosing random azimuths, avoiding localized optimal traps. The stochastic angle in Eq. (25) governs the trajectory followed by the agent in its search process. The SCSO algorithm achieves a balance between exploration and exploitation by adjusting the adaptable parameter vector r.

$$R = 2 \times \vec{r}_G \times rand(0,1) - \vec{r}_G \tag{26}$$

$\vec{r}_G$ linearly decreases from 2 to 0 as iterations progress. The modified depiction of sand cat locations is presented below (during both exploration and exploitation stages):

$$\vec{X}(t+1) = \begin{cases} \vec{r}.(\vec{X}_b(t) - rand(0,1).\vec{X}_c(t)) & |R| > 1 \\ \vec{X}_b(t) - \vec{r}.\vec{X}_b(t).cos(\theta) & |R| \leq 1 \end{cases} \tag{27}$$

In this algorithm, the exploration phase initiates an attack on the target when the absolute magnitude of R is less than or equal to 1. When $|R| > 1$, the exploration phase shifts to a global exploration mode, searching for potential solutions across a broader range. Each sand cat has a distinct exploration scope during the exploration stage, preventing the algorithm from getting trapped in locall optimal solutions. Algorithm 2 and Fig. 3 present the pseudocode and schematic structure of SCSO [45]. According to the flowchart and pseudocode of the algorithm, a brief, step-by-step overview of SCSO is as follows:

1. *Initialization*: Start with a population of potential solutions (sand cats) and set initial parameters.
2. *Position and Velocity Update*: Update the position and velocity of each sand cat based on its own experience and the experiences of others.
3. *Evaluation*: Assess the fitness of each sand cat using the objective function.
4. *Update Best Solutions*: Record the best positions found by each sand cat and update the global best position if necessary.
5. *Convergence Check*: Determine if the stopping criteria are met, such as reaching the maximum number of iterations or achieving a satisfactory solution.
6. *Output Best Solution*: Provide the best solution found as the final result.

Algorithm 2 Pseudocode of SCSO Algorithm

Start the population setup
Calculate the fitness metric
Begin the r; $\vec{r}_G$; R
while $(t \leq iter_{max})$ ***do***
for *each agent* ***do***
Find an arbitrary angle $(0 \circ \leq \theta \leq 360 \circ)$
if $(|R| \leq 1)$ ***then***
Update the search operative's location
else
Update the search operative's location
end if
end for
$t = t + 1$
end while

The run time of the SVSC, RFSC, CASC, and XGSC hybrid models developed by optimizing four models of SVR, RFR, CAT, and XGB by SCSO were 4.857,6.391, 9.497, and 5.104 s respectively.

5 Data collection and feature selection

Players' characteristics (*Height*, *Weight*, *Age*, …), performance (*Dribbling*, *Finishing*, *Shot power*, …), and popularity (which can be estimated through fans comments in social network applications) may be less or more influential on market value of players. Current literature extensively addresses the challenge of assessing key factors influencing football players' market value using feature selection techniques [3, 46–48]. Studies emphasize non-linear relationships, especially with age [14, 49, 50]. Lucifora and Simmons [51] found a positive impact of age on player salaries, peaking around 28 years but diminishing after that due to declining physical conditions. Others [52] explored how footedness affects salaries and how positions impact player earnings. Franck and Nuesch [53] highlighted the significance of player shooting attributes, particularly shots on goal, on market value.

The dataset utilized in this research is sourced from the FIFA 19 video game (as a simulator developed by EA Sports), along with real-world statistical reports extracted from [54], for which https://sofifa.com/?r=190075&set=true was the original reference. Sofifa.com is a prominent online database for football-related data, particularly focusing on player statistics within virtual football simulation games. The platform offers a wealth of information, including player attributes, performance metrics, popularity, and transfer market value across various seasons. The popularity of FIFA games provides an interesting parallel with real-world statistical reports in the field of football. Real-world statistics, including player performances, team dynamics, and strategic analyses, complement the virtual experience by offering insights into the broader football landscape. This broad dataset initially contained 53 features for 491 sample players related to the FIFA 19 (2018/2019) transfer market, which needed data engineering.

In the initial phase of dataset refinement, seven samples were excluded from the dataset due to missing feature values. Subsequently, the player's associated leagues were identified based on their respective club names. A predominant subset of 459 player samples was attributed to five top-ranked European leagues, namely Serie

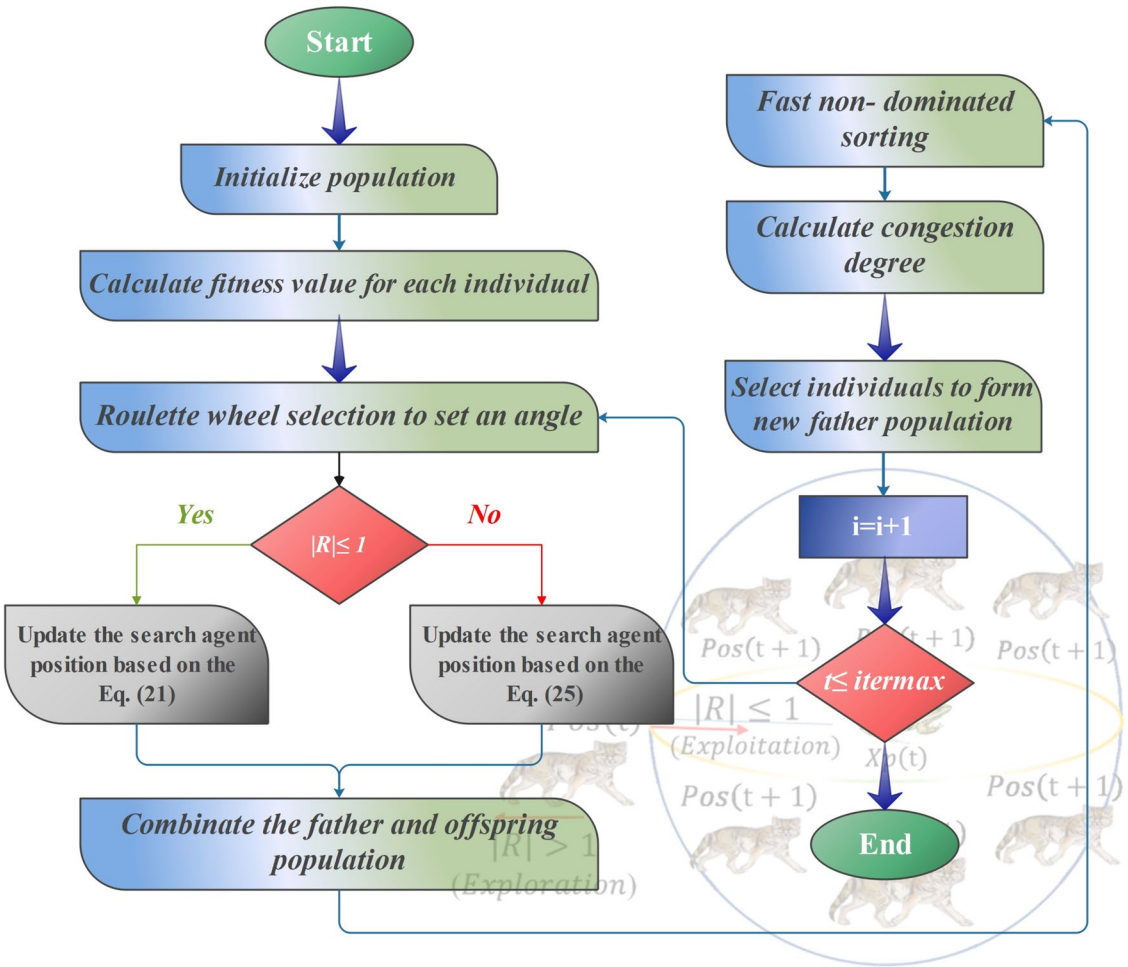


Fig. 3 Flowchart of Sand Cat Swarm Optimization (SCSO)

A (Italy), Premier League (England), League 1 (France), La Liga (Spain), and Bundesliga (Germany), while an additional 25 samples from less prominent leagues featuring players of lower value (considered outliers) were excluded. The dataset underwent a final refinement involving the removal of non-analytical feature columns, such as players' names and nationality. Moreover, certain nominal features, like preferred foot and weak foot, were appropriately labeled to align with the requisites of machine learning regression tasks. Finally, a cleaned dataset comprising 45 features (as listed in Table 1) for 459 sample players from five top-tier European leagues was provided for market value prediction using Python programming software.

Subsequently, SHAP (Shapley Additive Explanations) values for all four single models (SVR, RFR, XGB, and CAT) were utilized for feature selection by ranking the most important features in predicting football players' value. SHAP combines game theory and local explanations [55]. It simplifies inputs by mapping x to z via $x = h_x(z)$. The original model $f(x)$ is approximated using a linear function of binary variables based on z.

$$f(x) = g(x) = \varphi_0 + \sum_{i=1}^{M} \varphi_i z_i \tag{28}$$

Here, z is a binary vector with dimensions $\{0, 1\}^M$, M represents the number of input features. $\varphi_0$corresponds to $h_x(0)$, and φ_i represents the feature attribution value.

$$\varphi_i = \sum_{S \in F \setminus \{i\}} \frac{|S|!(M - |S|! - 1)!}{M!} [f_x(S \cup \{i\}) - f_x(S)] \tag{29}$$

$$f_x(S) = f\left(h_x^{-1}(z)\right) = E\left[f(x)|x_s\right] \tag{30}$$

Let F be the non-zero set of inputs in z, and S be the subset of F with the *ith* feature excluded. The unified measure of additive feature attributions termed the

Table 1 Description of the variables in the datase

NO.	Feature	Description
1	Age	The age of the player affects both the experience and future potential of players.
2	*Preferred Foot*	Dominant or more proficient foot that a soccer player uses for shooting, passing, and dribbling.
3	*International Reputation*	Globally perceived measure of trust, esteem, and recognition of a player shaped by achievements.
4	*Weak Foot*	The weakness of the player in using both legs in football reflects the level of inflexibility in the players.
5	*Skill Moves*	Techniques performed by players to outmaneuver opponents involve intricate ball control, dribbling, and feints.
6	*Height*	Height of the player affects the likelihood of scoring or preventing a goal.
7	*Weight*	The weight of the player affects the movement skills of the players.
8	*Crossing*	Technique where a player delivers the ball into the penalty area from the flanks
9	*Finishing*	player's ability to successfully score goals
10	*Heading Accuracy*	Player's proficiency in directing the ball with their head.
11	*Short Passing*	The number of passes to other players and the accuracy of passing.
12	*Volleys*	The technique is where a player strikes the ball while it is in the air without allowing it to touch the ground.
13	*Dribbling*	Skill that involves a player using controlled touches to maneuver the ball while on the move
14	*Curve*	Bending or swerving trajectory applied to the ball by the player during a shot or a pass.
15	*FK Accuracy*	Accuracy in taking free kicks.
16	*Long Passing*	The number of passes delivering the ball over a significant distance to a teammate.
17	*Ball Control*	player's skill in skilfully receiving, trapping, and manipulating the ball using various body parts.
18	*Acceleration*	How quickly a player can reach their top speed
19	*Sprint Speed*	Maximum velocity a player can achieve during a full-out sprint.
20	*Agility*	player's ability to change direction rapidly.
21	*Reactions*	player's quick responses to the movement of the ball, changes in the game situation, or the actions of opponents and teammates.
22	*Balance*	player's ability to maintain stability and control their body position during various movements.
23	*Shot Power*	strength with which a player strikes the ball during a shot on goal.
24	*Jumping*	Jumping ability of the player.
25	*Stamina*	player's overall ability to sustain physical effort and performance over an extended period of time.
26	*Strength*	player's physical power and ability to exert force against resistance.
27	*Long Shots*	The number of successful shots from a considerable distance away from the goal, often outside the penalty area.
28	*Aggression*	Player's assertiveness and determination in challenging for the ball.
29	*Interception*	Successfully blocks a pass or a ball played by the opposing team.
30	*Positioning*	Playing the position of the player.
31	*Vision*	player's ability to perceive and understand the unfolding dynamics of the game.

Table 1 (continued)

NO.	Feature	Description
32	*Penalties*	The number and accuracy of penalty kicks by a player.
33	*Composure*	player's ability to maintain calmness, control, and mental focus in high-pressure situations during a match.
34	*Marking*	The tactic of closely tracking and guarding an opponent to prevent them from receiving or playing the ball effectively.
35	*Standing Tackle*	Performance of player in standing tackles.
36	*Sliding Tackle*	Performance of player in sliding tackles.
37	*Games Played*	The number of games played by the player.
38	*Games Started*	The number of games started by the player.
39	*Minutes Played*	Playing time (minutes)for players.
40	*Goals*	The number of goals scored by the player.
41	*Assist*	Helping other players score a goal.
42	*Shots on Goal*	The number of shots of a player toward the goal.
43	*Shots*	The total number of shots by player.
44	*Yellow Card*	The number of yellow cards received by a player.
45	*Red Card*	The number of red cards received by a player.

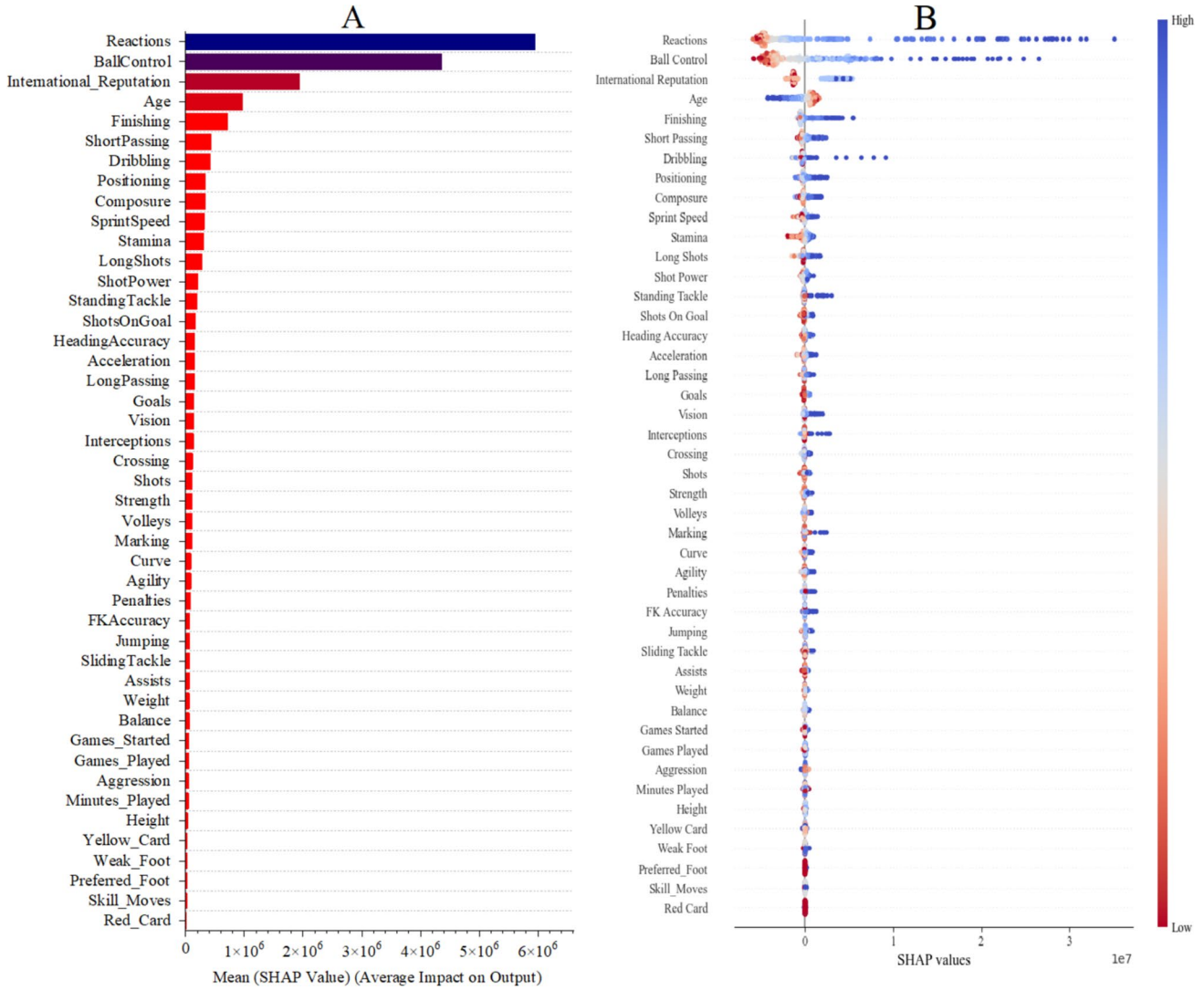


Fig. 4 SHAP feature-ranking plots of the RFR model for market value

SHAP value, is denoted as φ_i. SHAP is the only method possessing local accuracy, missingness, and consistency properties [56].

Calculating $E[f(x)|x_s]$ precisely is challenging, leading to the development of approximation methods such as *Tree SHAP*. *Tree SHAP* efficiently calculates $E[f(x)|x_s]$by utilizing decision tree structures. In the context of decision trees, T represents the number of trees in the ensemble, L represents the maximum number of leaves in any tree, M represents the number of input features, and D represents the maximum depth of a tree.

Without using *Tree SHAP*, the original complexity of calculating SHAP values is $O(TL2^M)$, The factor of 2 arises due to the interactions between the features for which the SHAP values are being computed. However, *Tree SHAP* reduces the complexity to $O(TLD^2)$, where the maximum tree depth D is used to limit the computations to a more efficient structure, providing a substantial computational advantage [57].

The input to *Tree SHAP* includes a trained tree-based model and input data X used for training (an $N \times M$ matrix with N instances and M features). This yields an $N \times M$ matrix of SHAP values, where each value indicates the contribution of the corresponding feature to the prediction of the instance. Figures 4, 5, 6 and 7 present SHAP feature-ranking in the case of each model, among which 20 top parameters were selected and utilized in the subsequent development of hybrid and ensemble models as input parameters.

According to Table 2, which reports a number of top-ranked features for well-known players in the dataset, their diverse skills can be interpreted as follows:

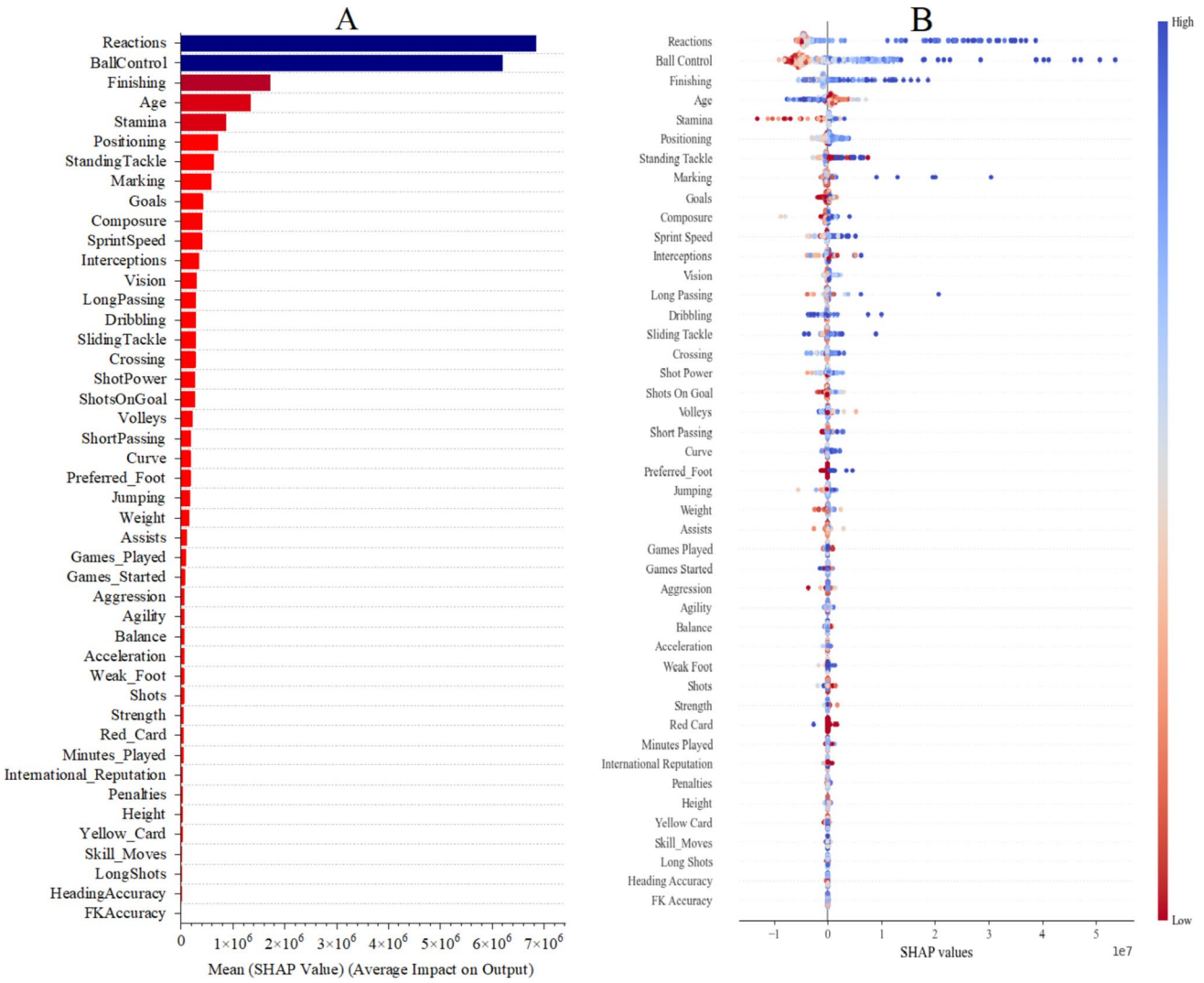


Fig. 5 SHAP feature-ranking plots of the SVR model for market value

- **Lionel Messi**, affiliated with Barcelona, stands out with exceptional reactions, ball control, and finishing, embodying the role of a prolific forward.
- **Cristiano Ronaldo**, representing Juventus, boasts a well-rounded skill set, excelling in reactions, ball control, and shot power, aligning with his forward position.
- **Neymar Jr.**, a key player for Paris Saint-Germain, is renowned for his skillful dribbling, solidifying his role as an offensive force.
- **Eden Hazard**, who moved to Real Madrid, showed finesse in ball control and composure, contributing as an attacking midfielder.
- **Kylian Mbappe**, a rising star at Paris Saint-Germain, distinguished himself with remarkable acceleration, sprint speed, and finishing, epitomizing the role of a dynamic forward.
- **Raheem Sterling**, a Manchester City player, is characterized by his Acceleration and composure, an influential winger in the offensive role.

6 Prediction evaluation, model innovation, and result analysis

6.1 Prediction performance measurement

In this study, the Coefficient of determination (R^2), Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Normalized Mean Square Error (NMSE), and

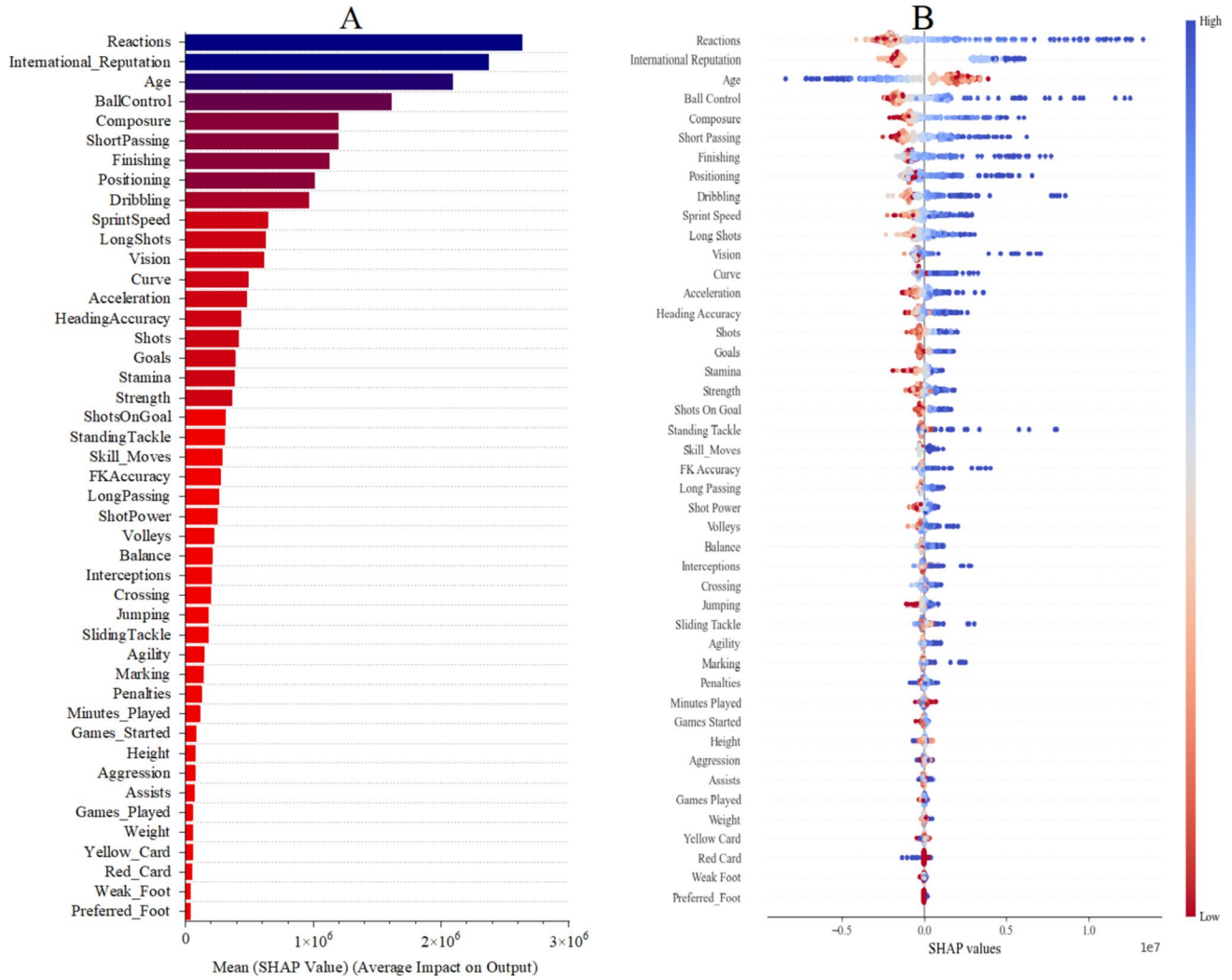


Fig. 6 SHAP feature-ranking plots of the CAT model for market value

Prediction Interval (PI) were utilized for evaluating the prediction performance of the developed models:

$$R^2 = \left(\frac{\sum_{i=1}^{n} (M_i - \bar{M})(P_i - \bar{P})}{\sqrt{\left[\sum_{i=1}^{n} (M_i - \bar{P})^2\right]\left[\sum_{i=1}^{n} (P_i - \bar{P})^2\right]}} \right)^2 \tag{31}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (M_i - P_i)^2} \tag{32}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n} (M_i - P_i)^2 \tag{33}$$

$$PI = \bar{x}_2 \pm t_{(\alpha/2,\ N-2)} * k^2 \tag{34}$$

$$NMSE = \frac{\sum_{i=1}^{n} (M_i - P_i)^2}{\sum_{i=1}^{n} (P_i)^2} \tag{35}$$

where M_i and $\bar{M}$ are real market values and an average of all real values, P_i and P are predicted market values and an average of all estimated values for n samples in the dataset.k^2 is the standardized error value that has been pooled across both groups, $\bar{x}_2$ are utilized for two distinct groups under

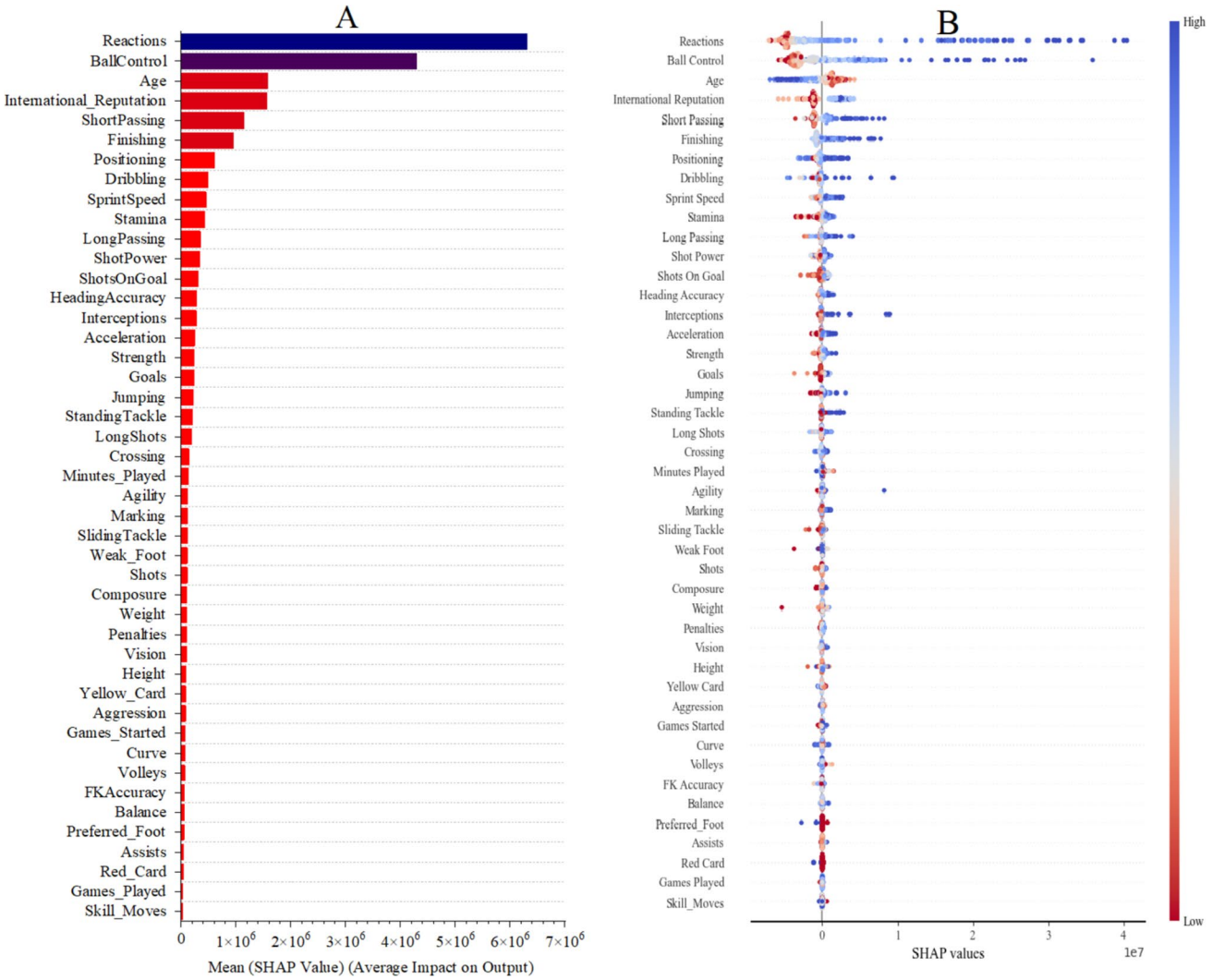


Fig. 7 SHAP feature-ranking plots of the XGB model for market value

comparison (predicted and real values) and $t_{(\alpha/2,\ N-2)}$is the t-value for the desired level of confidence (α) and the degrees of freedom ($N-2$).

6.2 Development of hybrid and ensemble prediction approaches

To develop hybrid prediction models, SCSO and HGSO optimizers were employed for fine-tuning hyperparameters of single models, as reported in Tables 3, 4, 5 and 6. Then, the convergence trend of the hybrid models is illustrated in Fig. 8, where it is visible that the error values of SVR-based models are two-fold higher than that of other models in both the initial and final point of 200 optimization iterations. Prediction models did not have a significant decrease in RMSE values after the 120th iteration. However, prediction by XGB-based models started with a higher error value, but optimization (especially by SCSO) led to enhanced estimation accuracy by six times. To conclude, according to this figure, developed models (except SVR-based models) had reasonable estimation based on time and cost-effectiveness.

To improve the reliability of approximations, it is essential to explore different prediction models and combine their outputs into an ensemble, where the final prediction is the average of individual model predictions. Since models vary in performance, a weighted average ensemble adjusts the infuluence of each model correlated

Table 2 Top-ranked feature values for some of the well-known players

Feature	Players					
	Lionel Messi	Cristiano Ronaldo	Neymar Jr	Eden Hazard	Kylian Mbappe	Raheem Sterling
	Clubs					
	Barcelona	Juventus	Paris Saint-Germain	Real Madrid	Paris Saint-Germain	Manchester City
Reactions	95	96	94	90	87	87
Ball Control	96	94	95	94	91	87
International Reputation	5	5	5	4	3	3
Age	31	33	26	27	19	23
Finishing	95	94	87	84	88	77
Composure	96	95	94	91	86	77
Dribbling	97	88	96	95	90	88
Stamina	72	88	81	83	83	78
Sprint Speed	86	91	90	88	96	92
Short Passing	90	81	84	89	82	84
Positioning	94	95	89	87	88	87
Shot Power	85	95	80	82	79	73
Heading Accuracy	70	89	62	61	77	38
Vision	94	82	87	89	82	77
Strength	59	79	49	66	71	52
Shots on Goal	89	65	26	43	68	39
Acceleration	91	89	94	94	96	95
Standing Tackle	28	31	24	27	34	58
Long shots	94	93	82	80	78	73
Interception	22	29	36	41	38	30

Table 3 RFR's hyperparameters optimization results

Hyperparameter	Models	
	RFHG	RFSC
n_estimators	1	3
max_depth	1	3
min_samples_split	2	3

Table 4 SVR's hyperparameters optimization results

Hyperparameter	Models	
	SVHG	SVSC
C	61.34910114	16.3254
Epsilon	0.01	0.01
Gama	1000	1292

Table 5 CAT's hyperparameters optimization results

Hyperparameter	Models	
	CAHG	CASC
bagging_temperature	0.01	0.00172165
learning_rate	0.328	0.00100351
random_strength	0.001	0.0028846
l2_leaf_reg	1	1
max_ctr_complexity	0.01	0.0382079
fold_len_multiplier	2	2
thread_count	1000	19

Table 6 XGB's hyperparameters optimization results

Hyperparameter	Models	
	XGHG	XGSC
n_estimators	1	2
max_depth	1	1
learning_rate	0.001	0.00211529
colsample_bytree	0.001	0.00175983
subsample	0.001	0.00236141
reg_alpha	0.001	0.00104718
reg_lambda	0.001	0.00184098

with its effectiveness, with more weight given to models that perform better. Performance metrics such as R^2 and RMSE are used to assign weights, with higher R^2 and lower RMSE indicating stronger models. These metrics are normalized and applied using the Dempster-Shafer Theory (DST) of evidence, which converts performance measures into probabilities known as basic probability assignments (BPAs). These BPAs are further transformed

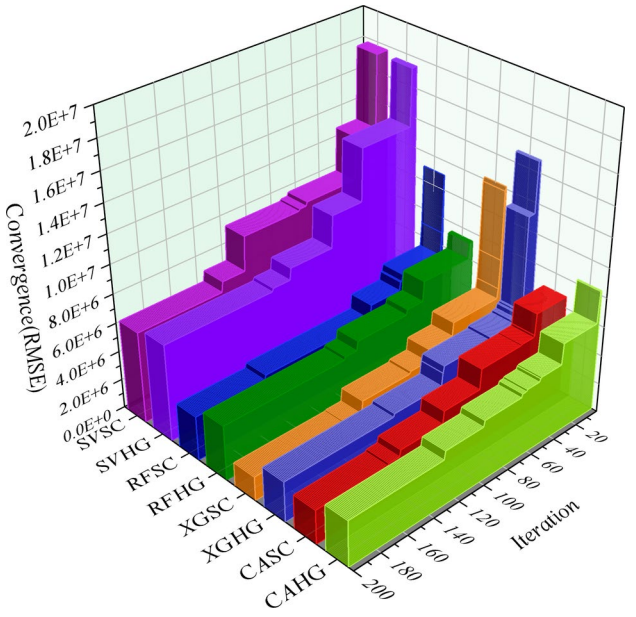


Fig. 8 Convergence pattern of hybrid models in market value prediction

into pignistic probabilities, which determine the weight each model contributes to the ensemble. Models with better performance receive higher weights, ensuring a more accurate and reliable final prediction.

All possible combinations of ensembles, including two-, three-, and four-model combinations, standalone models, and the full set of models, are evaluated. Statistical performance indices are computed normalized to obtain BPAs, and then DST is applied to calculate pignistic probabilities. The higher pignistic probability of the four models' ensemble indicated superior performance among the considered model combinations. Figure 9 presents the ensemble model development methodology for market value prediction in this study.

6.3 Prediction results and discussion

According to Table 7, R^2 values for all SVR-based and XGB-based models were the lowest and highest, and the optimization performance of SCSO was stronger than HGSO. As expected, the RSCX_SC ensemble model was the best predictor with a maximum R^2 value of 0.9931, XGSC and CASC with R^2 values of 0.9917 and 0.9830 came in the second and third ranking. According to comparative trends in metric values in Fig. 10, the prediction performance of hybrid forms of SVR was even weaker than RFR, CAT, and XGB single models (higher error values of RMSE, NMSE, and MAE values in Fig. 10A, C, and D, respectively, higher PI in Fig. 10E and lower R^2 in Fig. 10B).

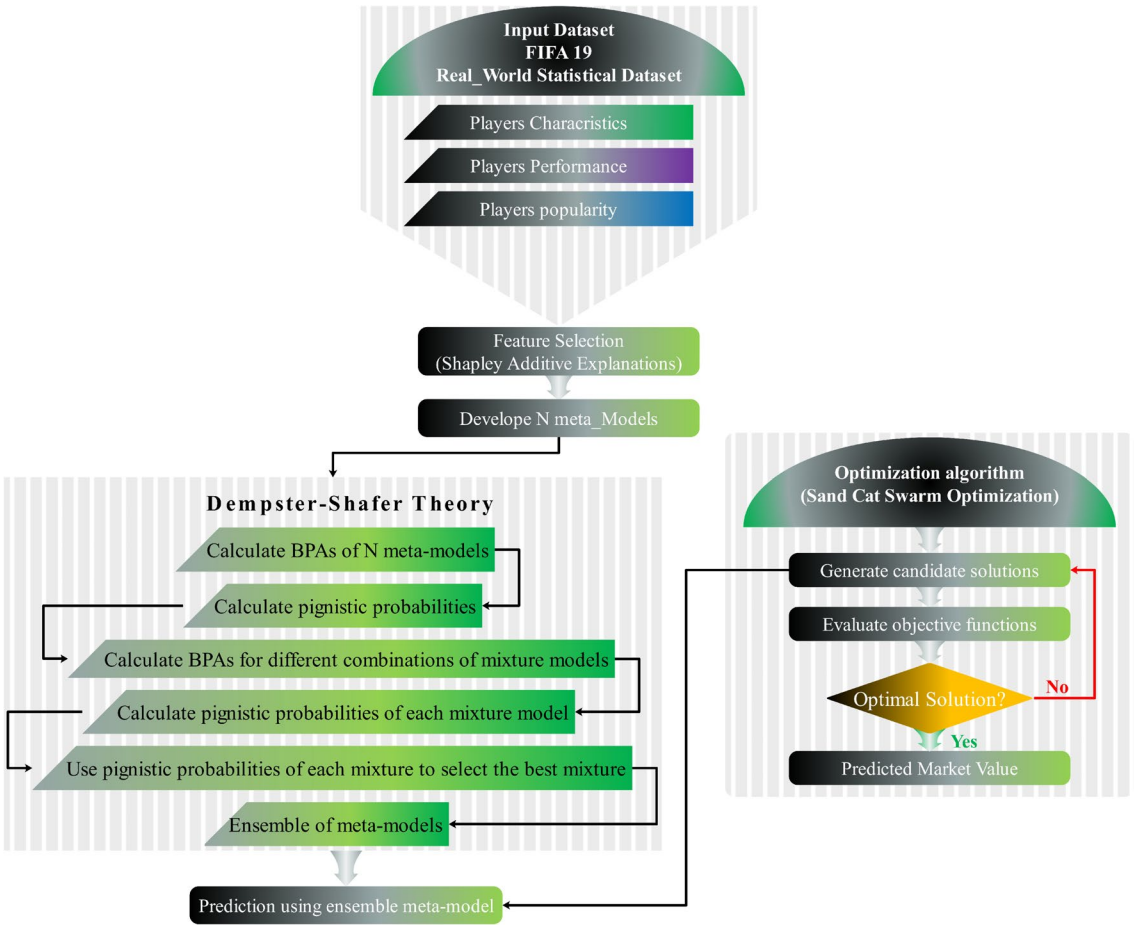


Fig. 9 Structure of the ensemble-based estimation of market value

Referring to the maximum, average, and minimum market values of all sample players in the dataset were 0.325, 17.3, and 119 million dollars, respectively; the minimum error value of 1.7 million dollars by XGSC was just 10% of the average market value and 1.4% of maximum value. RSCX_SC and CASC, with 2 (22% of average market value and 1.9% of maximum value) and 2.3 million dollars (13.3% of average market value and 1.9% of maximum value) misestimation, were the second and third most precise models. The high precision of XGSC and RSCX_SC was proved by a minimum PI value of 0.0480 and 0.0587, which means that the predicted values are more tightly clustered around the true values.

These comparisons indicated that such prediction models, especially the RSCX_SC ensemble model with more reliable accuracy, are the best alternatives for clubs' managers and stakeholders to rank the best decisions among players for the coming season based on the performance of players in the previous season. Also, with a simple estimation for all 11 players of a famous team (most preferred for clubs with a market value of players higher than average), managers can decide to concentrate more on which performance features of the players to select the ones with the most optimal salary.

Analyzing scatter plots and error plots concurrently, as highlighted in Fig. 11, offers an understanding of the prediction performance of models. In the scatter plots, the data points of XGB and CAT-based models had the nearest position to the diagonal line and fewer data points out of 15% over and underestimation (lower RMSE and higher R^2 values). A prominent concentration around the center line was related to RFSC, XGSC, and CASC hybrid models, while optimization did not enhance the performance of SVC that much, and it had noticeable errors, especially in predicting higher values, which may be more dangerous for well-known clubs with expensive players. Also, the error plots revealed that a majority of errors were related to positive values, which implies that club managers can safely rely on market values predicted by most optimal models (especially XGSC) with a low possibility of financial deficiencies.

Table 7 The result of developed models for SVR and RFR

Model		phase	Index values				
			RMSE	R^2	NMSE	MAE	PI
RFR	**Single**	Train	4.7E+06	0.9458	6.4E+10	3.2E+06	0.1223
		Validation	4.9E+06	0.9237	3.2E+11	2.8E+06	0.1699
		Test	3.1E+06	0.9511	1.3E+11	2.2E+06	0.1479
		Total	4.5E+06	0.9427	4.2E+10	3.0E+06	0.1318
RFHG	**Hybrid**	Train	3.8E+06	0.9673	4.4E+10	2.3E+06	0.1003
		Validation	4.3E+06	0.9456	2.5E+11	2.4E+06	0.1492
		Test	2.7E+06	0.9657	9.7E+10	1.9E+06	0.1276
		Total	3.8E+06	0.9638	2.9E+10	2.2E+06	0.1099
RFSC	**Hybrid**	Train	3.0E+06	0.9794	2.7E+10	1.9E+06	0.0788
		Validation	3.7E+06	0.9653	1.9E+11	2.1E+06	0.1272
		Test	2.3E+06	0.9732	7.5E+10	1.7E+06	0.1120
		Total	3.0E+06	0.9760	1.9E+10	1.9E+06	0.0886
SVR	**Single**	Train	8.9E+06	0.9240	2.3E+11	4.9E+06	0.2337
		Validation	4.9E+06	0.9223	3.3E+11	2.9E+06	0.1727
		Test	4.4E+06	0.9041	2.7E+11	2.4E+06	0.2158
		Total	7.8E+06	0.9206	1.3E+11	4.2E+06	0.2315
SVHG	**Hybrid**	Train	7.5E+06	0.9339	1.7E+11	4.4E+06	0.1975
		Validation	4.6E+06	0.9373	2.9E+11	2.7E+06	0.1591
		Test	4.0E+06	0.9111	2.2E+11	2.3E+06	0.1946
		Total	6.7E+06	0.9327	9.3E+10	3.8E+06	0.1973
SVSC	**Hybrid**	Train	7.1E+06	0.9442	1.5E+11	4.1E+06	0.1866
		Validation	3.9E+06	0.9374	2.1E+11	2.7E+06	0.1357
		Test	4.0E+06	0.9225	2.2E+11	2.5E+06	0.1958
		Total	6.3E+06	0.9415	8.3E+10	3.6E+06	0.1859
CAT	**Single**	Train	3.7E+06	0.9650	3.9E+10	2.7E+06	0.0952
		Validation	4.1E+06	0.9377	2.3E+11	2.4E+06	0.1424
		Test	2.4E+06	0.9693	8.0E+10	1.7E+06	0.1159
		Total	3.6E+06	0.9605	2.6E+10	2.5E+06	0.1041
CAHG	**Hybrid**	Train	3.1E+06	0.9756	2.8E+10	2.7E+06	0.0801
		Validation	4.2E+06	0.9659	2.4E+11	2.5E+06	0.1452
		Test	2.3E+06	0.9746	7.1E+10	1.8E+06	0.1093
		Total	3.2E+06	0.9690	3.2E+06	2.5E+06	0.0925
CASC	**Hybrid**	Train	2.4E+06	0.9847	1.7E+10	6.6E+05	0.0628
		Validation	2.4E+06	0.9683	8.1E+10	1.5E+06	0.0841
		Test	1.8E+06	0.9783	4.5E+10	1.3E+06	0.0866
		Total	2.3E+06	0.9830	1.1E+10	8.9E+05	0.0680
XGB	**Single**	Train	3.1E+06	0.9739	2.9E+10	2.4E+06	0.0811
		Validation	4.0E+06	0.9663	2.2E+11	2.1E+06	0.1385
		Test	1.7E+06	0.9826	4.0E+10	1.4E+06	0.0818
		Total	3.1E+06	0.9700	2.0E+10	2.2E+06	0.0904
XGHG	**Hybrid**	Train	2.6E+06	0.9846	2.1E+10	1.8E+06	0.0686
		Validation	3.7E+06	0.9768	1.9E+11	1.9E+06	0.1284
		Test	2.2E+06	0.9796	9.8E-01	1.7E+06	0.1065
		Total	2.8E+06	0.9808	1.6E+10	1.8E+06	0.0807
XGSC	**Hybrid**	Train	1.6E+06	0.9935	7.2E+09	1.2E+06	0.0404
		Validation	2.3E+06	0.9833	7.2E+10	1.3E+06	0.0792
		Test	1.3E+06	0.9894	2.3E+10	9.9E+05	0.0616
		Total	1.7E+06	0.9917	5.7E+09	1.2E+06	0.0480
RSCX_SC	**Ensemble**	Train	2.0E+06	0.9945	1.2E+10	1.2E+06	0.0520

Table 7 (continued)

Model	phase	Index values				
		RMSE	R^2	NMSE	MAE	PI
	Validation	2.5E+06	0.9850	8.4E+10	1.4E+06	0.0852
	Test	1.6E+06	0.9905	3.4E+10	1.1E+06	0.0749
	Total	2.0E+06	0.9931	8.5E+09	1.2E+06	0.0587

Figure 12 illustrates the sequential pattern of prediction errors across different models, highlighting the improvements in estimation accuracy achieved by employing hybrid and ensemble methods compared to traditional single models. The figure clearly shows that the error ranges of all hybrid models are lower than those of their single-model counterparts, particularly for the XGB and CAT-based models. This reduction in error demonstrates the enhanced performance and accuracy of the hybrid approaches, reinforcing their effectiveness in improving predictions compared to traditional models. For managerial decisions to buy players in the next season based on the performance of the players in the current season, it is less risky to have overestimations than underestimations, which was clearly obvious in error values of CASC, XGHG, and XGSC hybrid and RSCX_SC ensemble models. Moreover, the ensemble method (RSCX_SC model) not only lowered the error range but also had less outliers.

In this section, the Taylor plot in Fig. 13 represents a comprehensive comparison of the prediction performance of single, hybrid, and ensemble models. The measured value is the benchmark and position of each model's performance indicator (colored circles) in comparison with the real value is determined by Correlation Coefficient, Standard deviation, and RMSE values. The standard deviation of predicted values by SVR-based models was higher than that of RFR-based models, and optimization algorithms were not that successful in enhancing their performance. XGSC had the nearest position to the measured value, and CASC and RSCX_SC came in the subsequent ranking. It is worthwhile to note that RSCX_SC prediction results were the most reliable models even for data with special classification of players in groups with the same playing position or club, which enhances the accuracy of estimations by managers to select the most optimal player in each position with reasonable values.

6.4 Comparative analysis with previous studies

As reported in Section 1, there are some researches worked on the market value of football players. In this section, a comparison between the metric results of the best models of this study (XGSC and RSCX_SC) and previous literature was conducted to prove the high prediction accuracy of developed models. Based on reports in Table 8, the results of XGSC and RSCX_SC were 29–45% better than the prediction results of best-developed models in previous literature.

6.5 Sensitivity analysis

All sensitivity analysis methods share the common feature of adjusting parameters across model runs $j = 1,2,3,\ldots,N$, referred as the domain of model run. Fourier Amplitude Sensitivity Testing (FAST) uses Fourier series to estimate model output variance and sensitivity by varying parameters at distinct frequencies, linking Fourier coefficients to parameter effects. The process of generating this parameter set involves selecting suitable frequencies, creating a value set S with a uniform distribution between -0.5 and 0.5, and transforming S into the actual Fourier parameters [58].

As the number of parameters increases, the necessary number of model runs [59] also rises to ensure the independence of frequencies. Hence, the chosen frequencies (f_i) and the required sampling size N are contingent upon the number of parameters n (where $i = 1,\ldots,n$). Subsequently, the selected frequencies, along with the model run index, are transformed into a supporting variable $S(j,i)$ and then to actual Fourier parameters $\theta\ (j,\ i)$.

$$S(j,i) = \arcsin(\sin\{f_i(\pi\ /N)[(2j - N - 1)/2]\})/\pi \tag{36}$$

$$\theta\ (j,i) = F_i^{-1}(S(j,i) + 0.5) \tag{37}$$

where, F_i^{-1} is the inverse of the cumulative density function for $i - th$ parameter.

At each time step t, the model outputs $M = y(j,t)$ were subjected to a fast Fourier transform, producing a power spectrum. The variance S_i^2, explainable by a specific parameter i, was computed by summing the power in the spectrum for frequencies $f(i), 2f(i), 3f(i), 4f(i)$ (for further clarification on the use of higher-order frequencies up to 4, refer to [59]). The total variance (S^2) was computed as the sum of the power spectrum across all frequencies, with parameter sensitivity (S_i^2/S^2) reflecting the ratio of partial to total variance. Sensitivity analysis revealed that

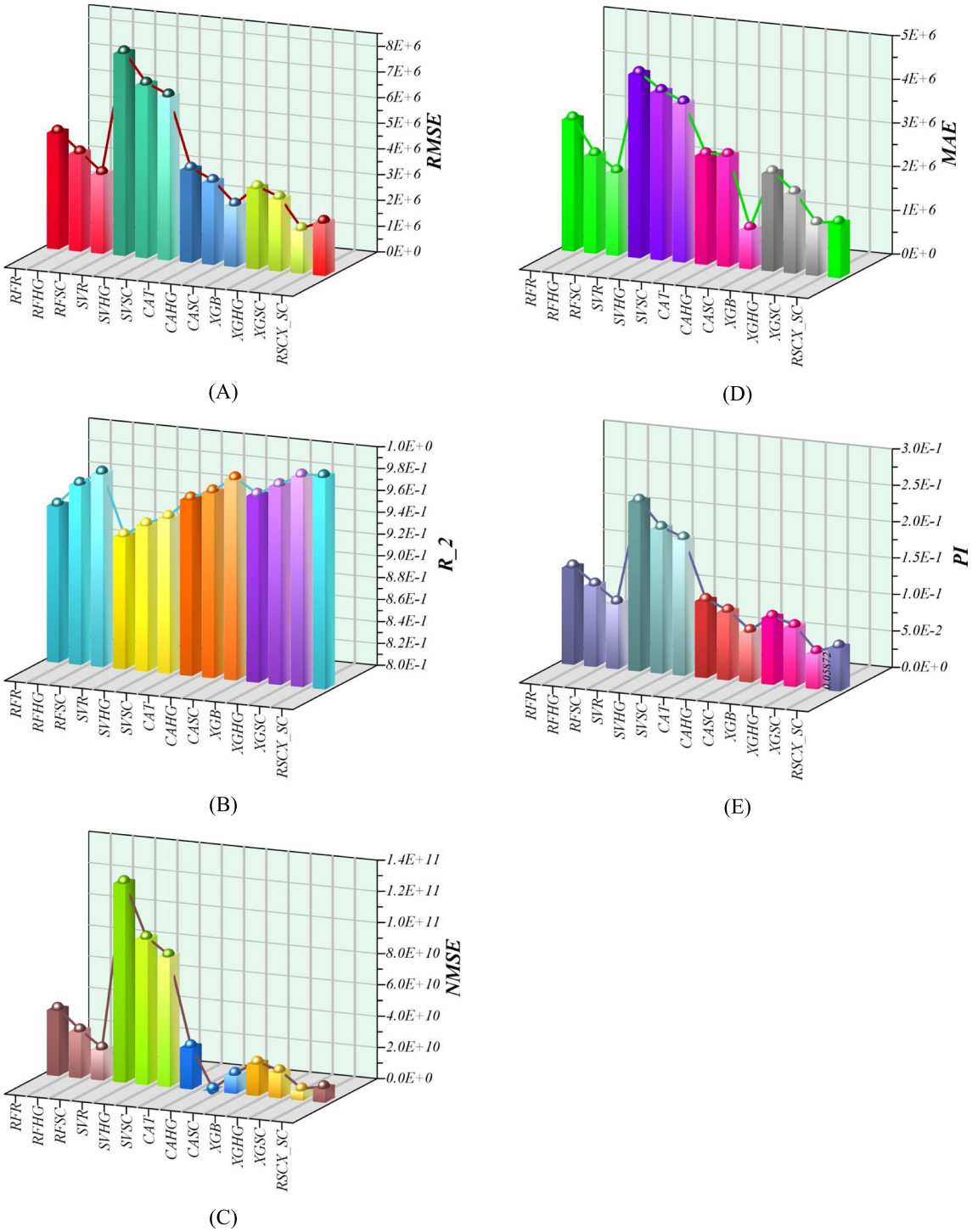


Fig. 10 Comparison of metric results for developed models in each version of selected features

Dribbling, Ball Control, and Standing Tackle were the most influential independent factors affecting player market value. Total sensitivity (STSTST) summed first-order effects and interactions, showing similar contributions across all 20 parameters, indicating their collective impact on player salary (Fig. 14).

7 Conclusion

The integration of machine learning methodologies into the analysis of football player data has become a turning point in predicting player values within the dynamic football market. The complexity of player datasets,

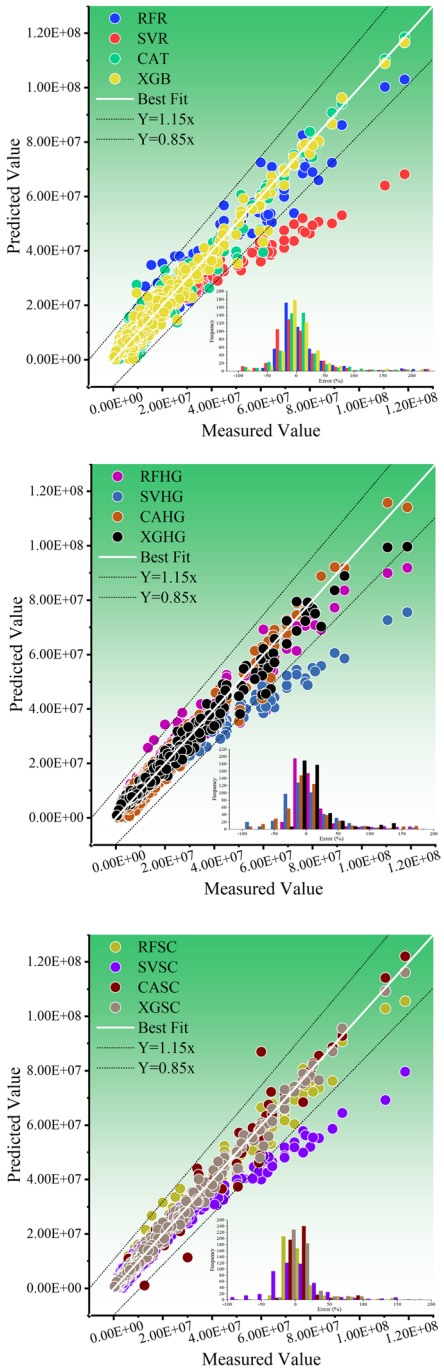


Fig. 11 Prediction performance of selected models

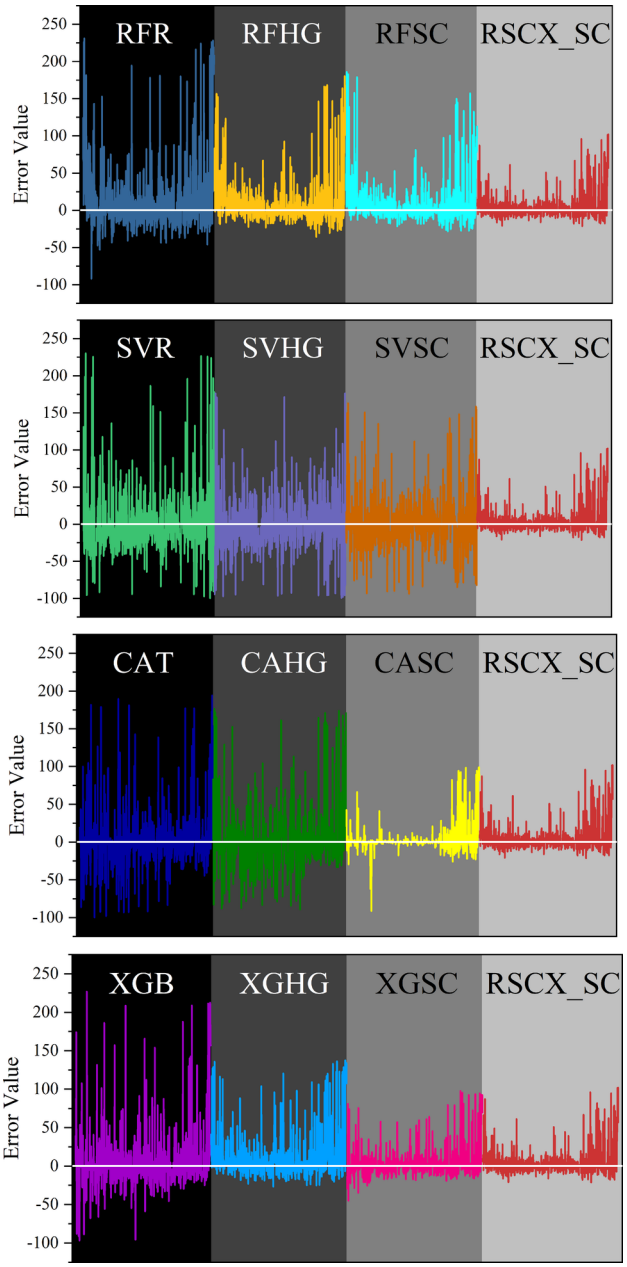


Fig. 12 Comparative prediction error for single, hybrid, and ensemble models

encompassing various dimensions such as performance metrics, Players' characteristics, performance, and popularity, presents both challenges and opportunities. Utilizing Machine Learning (ML) models, ranging from traditional regression techniques to advanced hybrid and ensemble methods, provides a systematic approach to extracting actionable insights from this intricate web of information. The predictive task, involving estimating a player's market value based on multiple input features, requires exploration of complex interactions and nonlinear relationships within the data. Despite challenges related to overfitting and computational complexity due to abundant dimensionality, the application of feature-selecting tools and sensitivity analysis tools becomes crucial. This study specifically examined the importance of 45 features in the dataset (FIFA 19 and Real-world Statistical Datasets) in market value prediction by Support Vector Regression (SVR), Random Forest Regression (RFR),

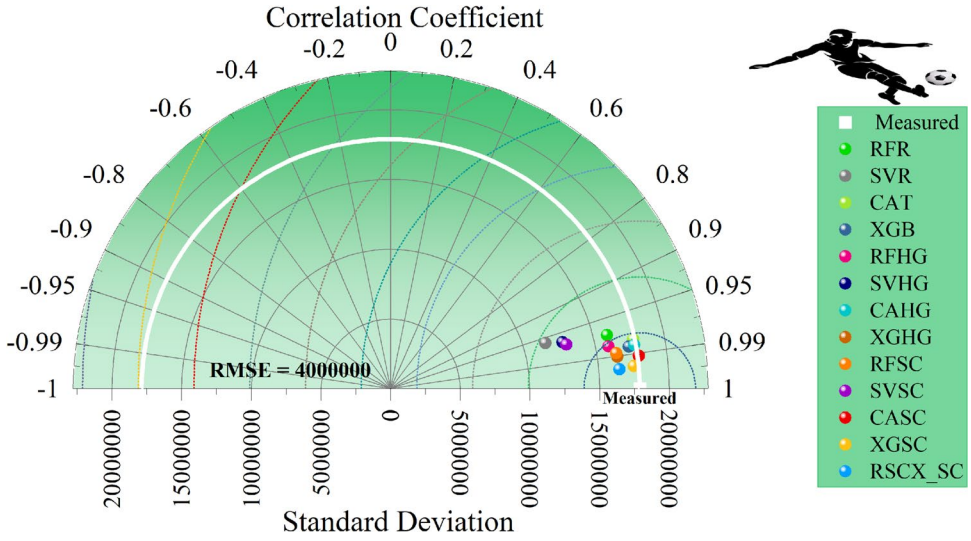


Fig. 13 Taylor plot for overall comparison between selected predictors

Table 8 Comparative analysis of previous studies

NO.	reference	Model	R^2	Differences with results of this study (XGSC and RSCX_SC)
1	[14]	PSO-SVR	0.74	−34%
2	[58]	xgbTree	0.77	−29%
3	[24]	RFR	0.95	−45%

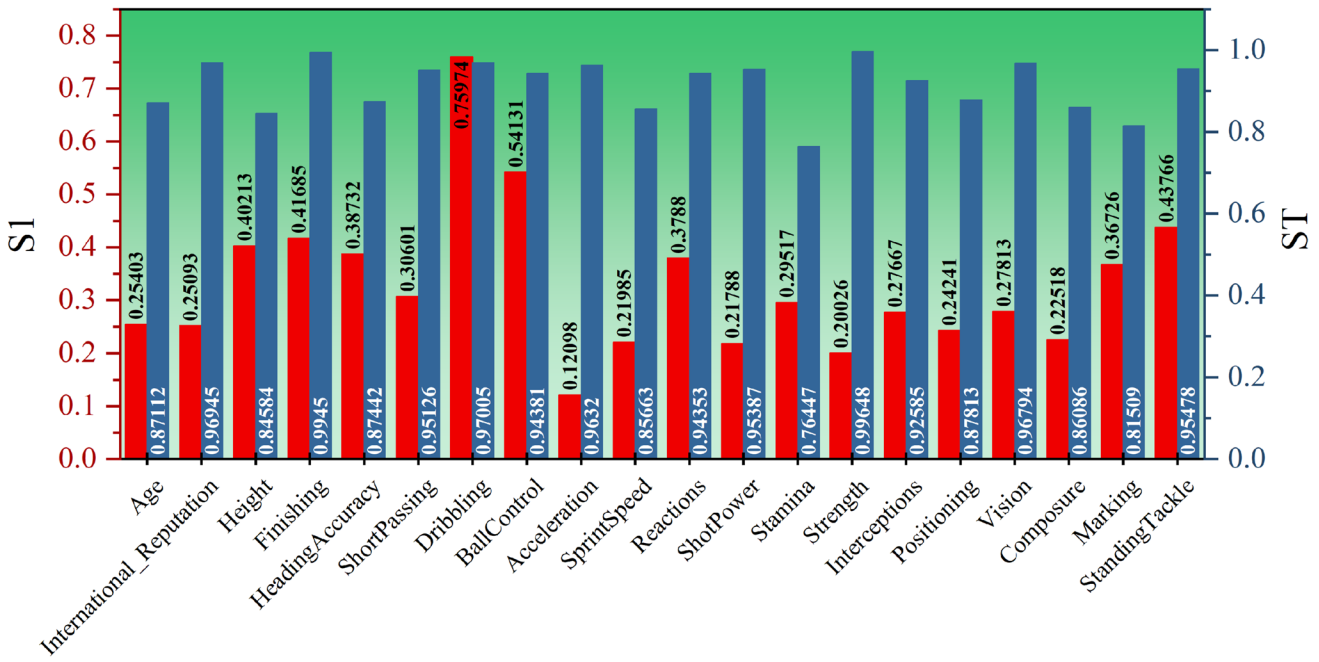


Fig. 14 The results of Fourier Amplitude Sensitivity Testing

Extreme Gradient Boosting (XGB), and Categorical Boosting (CAT) utilizing Shapley additive explanations (SHAP). Then, it introduced the most relevant features (20 out of 45) according to the SHAP importance ranking. The models were optimized with two meta-heuristic algorithms, and an ensemble model was developed by

Dempster Shafer's theory to address overfitting and generalization issues.

Based on the prediction results, SVR-based models had the lowest R^2 values, while XGB-based models achieved the highest. The SCSO optimization method performed better than HGSO, with the RSCX_SC ensemble prediction approach emerging as the top predictor with an R^2 of 0.9931. XGSC and CASC models followed with R^2 values of 0.9917 and 0.9830. XGSC and RSXC_SC as best performers are beneficial for optimizing player salaries for all 11 players in a team with around 19 and 22 million dollars cumulative error, especially for prominent teams with market values above average and high range of funds. It is noteworthy to highlight that the predictive outcomes of RSCX_SC were consistently the most dependable models, even when applied to data involving the distinct classification of players into groups based on their playing position or club affiliation. The application of machine learning in predicting player market values brings advantages such as heightened predictive accuracy, optimized transfer strategies, and the potential to uncover market inefficiencies.

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Declarations

Human participants and/or animals Not applicable.

Ethical approval Not applicable.

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